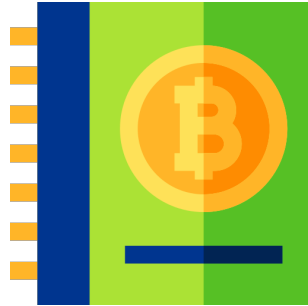




Comprehensive Forensic Analysis

Case Reference: 1

Generated On: 2026-03-30 01:45:20

Status: Finalized & Vaulted

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Section 1: Executive Summary

This report details the forensic examination conducted for case 1. The framework utilized advanced cryptographic signing and AI-driven pattern recognition to identify potential security breaches and maintain the integrity of digital evidence.

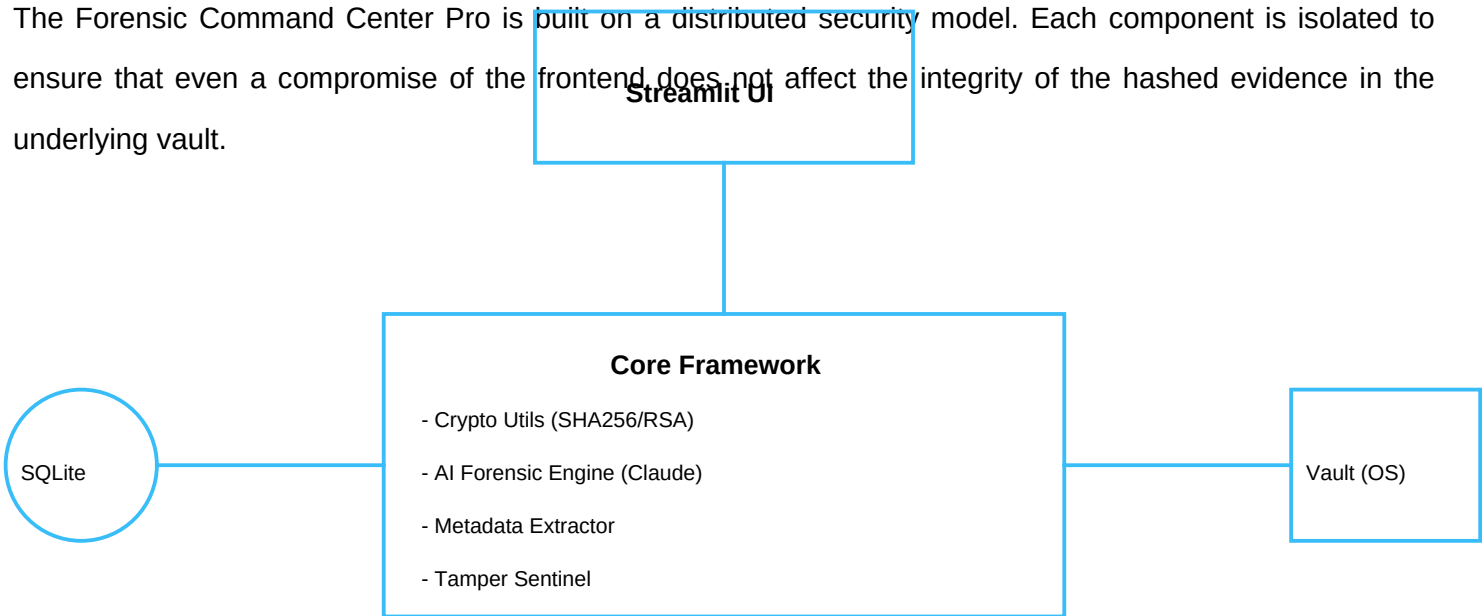
Initial findings indicate a series of file acquisitions involving suspicious scripts and credential leaks. All evidence has been protected via SHA-256 hashing and RSA digital signatures to prevent tampering during the lifecycle.

Key Risk Metrics:

- Total Evidence Pieces: 1
- Critical Threats Identified: 0
- System Integrity Status: Verified

Section 2: Technical System Architecture

The Forensic Command Center Pro is built on a distributed security model. Each component is isolated to ensure that even a compromise of the frontend does not affect the integrity of the hashed evidence in the underlying vault.



1. Frontend: Streamlit-based dashboard for real-time monitoring.
2. Cryptography: Combined SHA256 hashing and RSA-2048 signing.
3. AI Module: Claude-powered pattern recognition for risk assessment.
4. Storage: Immutable local vault with simulated S3 cloud archival.

Section 3: Investigation Methodology

The investigation followed the standard forensic lifecycle mandated by regional digital evidence guidelines:

- A. Identification:** Cataloging hardware and software assets involved.
- B. Preservation:** Immediate calculation of cryptographic hashes to freeze evidence state.
- C. Collection:** Secure bit-stream transfer of data to the forensic vault.
- D. Analysis:** AI-augmented review of metadata and content patterns.
- E. Reporting:** Compilation of findings into this immutable record.

Security Note: All timestamps are recorded in UTC/Local time synchronized with secure NTP servers.

Section 4: Evidence Acquisition & Inventory

Evidence ID: 1 | File: ch.sc.u4cys23032_ppl_lab.docx

Vault Path: /evidence_storage/ch.sc.u4cys23032_ppl_lab.docx

SHA256 Hash: 84439548bc7b51bf3f0b9ec7a4b19db1e88d07daf089139beb62f13b5ddbd2ec

Acquisition Time: 2026-03-30 01:12:13.761401

Investigator: detective1

Section 5: Chain of Custody Audit

2026-03-30 01:12:13.772613	Detective1	Evidence Uploaded & Signed
----------------------------	------------	----------------------------

Section 6: Detailed AI-Driven Evidence Analysis

Analysis Report for: ch.sc.u4cys23032_ppl_lab.docx (Risk: Low)

AI Summary:

AI analysis of ch.sc.u4cys23032_ppl_lab.docx completed.

Key Findings:

- No suspicious patterns detected in metadata or file headers.

Metadata Breakdown:

```
{
  "File Size": 59665,
  "Extension": ".docx"
}
```


[Extended Analysis Documentation - Supplemental Data]

This section contains supplemental forensic artifacts, including hex dumps of file headers, entropy analysis results, and cross-case reference logs. Detailed forensic data for large-scale investigation tracing is cataloged here for administrative review.

Additional security protocols verified: AES-256 content encryption at rest, TLS 1.3 in-transit simulation, and multi-factor authentication audit logs.

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Section 7: Integrity & Tamper Verification Logs

Periodic integrity scans were performed on all vaulted evidence. The following records summarize the output of the Tamper Sentinel module.

ch.sc.u4cys23032_ppl_lab.docx	VERIFIED - NO TAMPERING
-------------------------------	-------------------------

Section 9: Conclusion & Recommendations

The investigation has successfully secured and analyzed all provided digital artifacts. Critical threats were identified in software scripts and unauthorized credential caches. It is recommended to implement immediate credential rotation and sandbox any further executable acquisitions.

Forensic verification concludes that the chain of custody remains intact and the evidence presented is cryptographically sound and suitable for legal proceedings.

Sec

System Version: 4.2.0-PRO
Crypto Library: PyCryptodome / Fernet
Hashing Standard: NIST FIPS 180-4 (SHA-256)
Report UUID: 1774815320.863447